SWE 4634 **Network Security Lab** **Credit: 0.75**

Sessional works based on SWE 4633.

SWE 4636 **Computer Graphics and Multimedia Lab** **Credit: 0.75**

Sessional works based on SWE 4635.

Seventh Semester

Hum 4747 **Legal Issues and Cyber Law** **Credit: 3.00**

Introduction to legal aspects, Jurisdiction, Intellectual property laws (Copyright, patent, trademark, etc.), Contracts and licenses, Privacy in the workplace, Trade secrets and non-disclosure agreement;

Cyber laws and rights in today's digital age ICT Act; Information Warfare, computer crime and information terrorism; Threats to information resources, including military and economic espionage, communications eavesdropping, computer break-ins, denial-of-service, destruction and modification of data, distortion and fabrication of information, forgery, Digital Forensics, IT and the Legal Profession, Policing the Internet, Cyber Constitutionalism, Cyber Speech, Cyber Privacy, Cyber Defamation and Conflicts.

Recommended Texts:

1. C. Reed, *Computer Law*, 7th ed. Oxford University Press, 2012
2. D. S. Wall, *The Internet: Law and Society*, 1st ed., Y. Akdeniz and C. Walker, Eds. Longman, 2001
3. D. I. Bainbridge, *Introduction to Computer Law*, 5th ed. Longman Pub Group, 2004

SWE 4701 **Software Metrics and Process** **Credit: 3.00**

Definition of software measurement and metrics; The basics of measurement: Property-oriented measurement, Meaningfulness in measurement, Measurement quality, Measurement process, Scale, Measurement validation, Object-oriented measurement, Subjectdomain-oriented measurement; Goal-based framework for software measurement: Software measure classification, Goal-based paradigms, Case studies, Empirical investigation, Measuring internal product attributes: size, Structure: Software structural measurement, Controlflow structure, Cyclomatic complexity, Data flow and data structure attributes, Architectural measurement; Software Process Metrics, Testing

Metrics, Quality Metrics, Usability Metrics, Software Maintainability Metrics, Software cost model, COCOMO and COCOMO II, Measuring external product attributes: quality; Measuring software reliability: Software reliability models and metrics.

Recommended Texts:

1. N. Fenton and J. Bieman, *Software Metrics: A Rigorous and Practical Approach*, 3rd ed. CRC Press, Inc., 2014. [Online]. Available: `https://dl.acm.org/doi/10.5555/2700539`
2. B. A. Kitchenham, *Software Metrics: Measurement for Software Process Improvement*. Blackwell Publishers, Inc., 1996

CSE 4714 **Technical Report Writing** **Credit: 0.75**

Issues of technical writing and effective oral presentation in Computer Science and Engineering; Writing styles of definitions, propositions, theorems and proofs; Preparation of reports, research papers, theses and books: abstract, preface, contents, bibliography and index; Writing of book reviews and referee reports; Writing tools: LATEX; Diagram drawing software; presentation tools; Definition of plagiarism; Types of plagiarism; How to detect plagiarism; Plagiarism and world wide web; How to avoid plagiarism.

Recommended Texts:

1. B. Eunson, *Writing and Presenting Reports*, 1st ed. Wiley, 1994
2. R. P. Clark, *Writing Tools: 55 Essential Strategies for Every Writer*, 1st ed. Little, Brown Spark, 2008
3. H. Hering, *How to Write Technical and Scientific Reports: Understandable Structure, Good Design, Convincing Presentation*, 3rd ed. Springer, 2025
4. L. Lamport, *LaTeX: A Document Preparation System*, 2nd ed. Addison-Wesley Professional, 1994

SWE 4790 **Internship** **Credit: 9.00**

The student will work full-time as an Intern in a particular company for a period of 5/6 months. The students will join the company for Internship just after the 6th Semester final examination and will resume their classes after the Mid Semester Examination of the Winter Semester of the next academic year.